Facial Emotion Recognition

https://github.com/aiyshwary/Facial-Emotion-Detection

OVERVIEW

A comparative study of machine learning and deep learning approaches for recognizing facial emotions f
CNN, SVM, Random Forest, and Decision Tree models across multiple datasets.

IMPACT

Implemented and benchmarked four ML/DL models for 7-class facial emotion classification. CNN achieve
approximately 75.5% accuracy, outperforming all traditional models on unseen data.

KEY DETAILS

1. Trained on icml_face_data and Kaggle FER datasets with 7 emotion classes (happy, sad, angry, fear,
2. CNN with data augmentation and stratified cross-validation achieved approximately 75.5% validation a
3. Random Forest reduced overfitting vs Decision Tree, achieving approximately 89.2% on ICML dataset
4. SVM with PCA preprocessing achieved approximately 55% accuracy — lowest among all models on h
5. Concluded CNNs are best suited for FER due to hierarchical spatial feature extraction.
6. CNN architecture: 4 convolutional blocks (64-128-256-512 filters) with BatchNormalization, He Normal
   Dense(512) to Dense(7, softmax) head totaling ~7.1M parameters.
7. Input images are 48x48 grayscale pixels normalized to [0,1]; augmentation via Keras ImageDataGene
   width/height shift, shear, zoom (0.15), and horizontal flip.
8. Training used Adam optimizer with EarlyStopping (patience=11), ReduceLROnPlateau (factor=0.5, pa
   stratified cross-validation with batch size 32.